NeuroBridge-S4 Methodology Description

Small-N graph-informed adaptation management for Artemis II proxy data

Methodological Thesis

NeuroBridge-S4 is a small-N human adaptation management methodology for the statistical structure of Artemis II human research: four deeply characterized individuals, multimodal measurements, limited sampling, and a need for operationally meaningful interpretation. It does not force population-level inference onto n=4. It treats each crew member as a longitudinal biological system, uses real terrestrial data as structured reference space, and translates cross-system coherence into monitoring priorities.

Core value: turn rare, high-dimensional human spaceflight-like data into individualized adaptation profiles that connect biochemistry, physiology, neurocognition, evidence-based psychology, recovery, and HRP decision support.

Architecture: Five Linked Layers

Layer	Role	Output for HRP
1. Control Ladder	Use real proxy datasets as progressively relevant reference spaces: population, health-filtered cohort, longitudinal physiology, spaceflight analog, individual baseline.	Transparent reference context; no claim that one dataset substitutes for Artemis II.
2. Individual baseline logic	Analyze four people as four biological trajectories, not as an underpowered group average.	Pre/exposure/post/recovery deltas; baseline imbalance handling.
3. Domain compression	Map variables into biological domains: sleep/circadian, autonomic, stress biology, immune/inflammatory, metabolic, cardiovascular, cognition, psychological self-report.	Interpretable systems rather than scattered columns.
4. Graph coherence	Separate knowledge graph, observed proxy coherence graph, and decision graph.	Shows which systems move together and which interpretation is conceptual vs data-derived.
5. Management translation	Convert coherent patterns into monitoring priority, follow-up data streams, and countermeasure considerations.	Individual Adaptation Profile + Crew-Level Summary.

BACI: Biological Adaptation Coherence Index

BACI is a transparent signal-triage layer, not a clinical score. It elevates coordinated multi-domain adaptation patterns over isolated biomarker changes. A high BACI means multiple independent systems shift in a biologically compatible pattern and deserve prioritized review.

Component	Question answered
Deviation magnitude	How strongly does a person differ from their own baseline or reference space?
Domain breadth	How many independent biological systems are involved?
Temporal alignment	Do changes occur and recover together across mission/proxy phases?
Graph support	Are active domains connected in the knowledge graph and/or observed coherence graph?
Operational consequence	Which monitoring stream or countermeasure category logically follows?

Graph Representation: Why It Is Essential

A table fragments an astronaut into columns. A graph preserves the relational structure of human adaptation. NeuroBridge-S4 uses three graphs with distinct meanings: (1) a biological knowledge graph for literature-plausible pathways, (2) an observed proxy coherence graph computed from real proxy data, and (3) a decision graph that maps patterns to monitoring priorities. This separation prevents decorative overclaiming while giving HRP reviewers an interpretable system map.

Graph type	Nodes	Edges	Interpretation
Biological Knowledge Graph	Sleep, autonomic regulation, HPA-axis, immune signaling, metabolism, cognition, fatigue, emotional regulation, recovery.	Biologically plausible relationships.	Scientific map for interpreting multimodal data; not direct proof for a given astronaut.
Observed Proxy Coherence Graph	Measured variables or biological domains in real proxy data.	Correlation, co-deviation, or shared activation above threshold.	Shows which systems actually move together in the demonstration data.
Decision Graph	Observed pattern, biological meaning, functional implication, monitoring priority.	Translation logic.	Turns analysis into HRP-facing decision support.

Biochemical-Neurocognitive Layer

NeuroBridge-S4 explicitly connects biochemistry, neurobiology, cognitive function, and evidence-based psychology without diagnosing mental states from biomarkers. Acute cognitive-perceptual vulnerability is treated as a convergence context: sleep disruption, autonomic dysregulation, stress biology, immune-inflammatory shift, cognitive fatigue, and validated self-report together can indicate vulnerability to cognitive overload, emotional dysregulation, derealization-like detachment, or reduced recovery capacity. The output is monitoring priority, not diagnosis.

Sensitivity and Ablation

A 10/10 methodology must show that each layer adds value. The demonstration therefore includes BACI threshold sensitivity (0.75, 1.00, 1.25), an ablation view comparing raw biomarkers vs domain compression vs BACI vs graph-informed interpretation, and explicit separation of conceptual and data-derived graphs.

Dataset-Aware Demonstration Strategy

Actual Artemis II data are unavailable during the challenge. NeuroBridge-S4 therefore demonstrates the workflow on real proxy data, avoiding synthetic subjects. NHANES is used for reference-space construction and real n=4 pseudo-crew selection. Longitudinal wearable/sleep data, NASA Life Sciences Portal/UTMB bedrest, and ImmPort are defined as extension layers for full Artemis transfer.

Proxy layer	Role	Why it matters
NHANES	Health-filtered terrestrial reference; real n=4 pseudo-crew; biomarker deviation.	Shows reference-space and small-N profile construction on real noisy public data.
PhysioNet / wearable sleep-stress data	Longitudinal physiology, actigraphy, sleep, HR/HRV, stress/recovery.	Adds time dynamics and within-subject recovery modeling.
NASA Life Sciences Portal / UTMB bedrest	Small-N analog exposure and recovery.	Closest proxy for mission-like constraints and countermeasure imbalance.
ImmPort	Immune reference and stress-immune interpretation.	Supports Artemis Immune Biomarkers transfer.

What HRP Gets

Deliverable	Practical value
Individual Adaptation Profile	Which systems changed, whether the pattern is coherent, and what follow-up is highest value.
Crew-Level Summary	Shared vs divergent adaptation patterns across four people without averaging away individuals.
Monitoring Priority Map	Translates patterns into sleep/circadian, immune, cognitive, psychological, recovery, or workload review.
Data Gap Recommendations	Identifies which future measurements would most increase interpretability.

Boundary Conditions

NeuroBridge-S4 does not claim to diagnose, predict clinical outcomes, infer brain chemistry directly, or replace HRP clinical judgment. Its value is a transparent, reproducible, graph-informed decision-support methodology for rare, multimodal human spaceflight data.